Challenging Question Session I

Let P, Q, and R be three points in a plane such that P is origin and Q lie on the positive side of x axis and is 6 units away from origin. R lie on the negative side of y axis and is 6 units away from origin. A point M divides the line segment PQ in the ratio 2:1, A line which passes point M and is parallel to Y-Axis cuts the line segment RQ at point N, then which of the following options is/are all correct.

co-ordinate of point N is (4, −2)
- ◯ co-ordinate of point N is (4, −3)
- ◯ Equation of line QR is $\frac{y}{-6} + \frac{x}{6} = 1$
- ◯ Equation of line QR is Y = X − 6
- ◯ The Ratio of RN to NQ is 1 : 2
- ◯ The Area of Quadrilateral PMNR is 12
- ◯ The Area of Quadrilateral PMNR is 16

Solution:

As given in question Point P is $(0,0)$, Q is $(6,0)$ and R is $(0,-6)$.

Since M divides line segment PQ in ratio $2:1$, hence by applying Section Formula:

$$(x,y)=\left(\frac{mx_2+nx_1}{m+n},\frac{my_2+ny_1}{m+n}\right)$$

$$(x,y)=\left(\frac{2\times 6+1\times 0}{2+1},\frac{2\times 0+1\times 0}{2+1}\right)$$

$$(x,y)=(4,0)$$

Hence coordinate of M is (4,0)

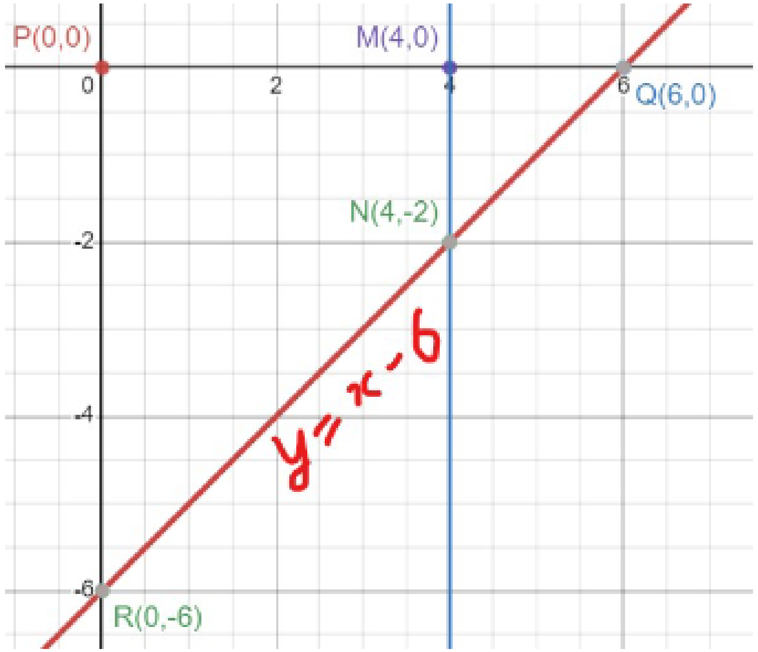


Figure 1

Equation of Line RQ can be found using two-point form. We'll get equation of RQ as $y = x - 6$. Therefore, Option(4) is correct.

Also, the same line can be written in the form of Intercept form. with $x - intercept$ as 6 and $y - intercept$ as -6. Therefore writing it in Intercept form:

$$\frac{x}{6} + \frac{y}{-6} = 1$$

Therefore Option(3) is correct.

As given in the question, the intersection point of Line $x = 4$ and $y = x - 6$ is (4,-2) which is point N.

Therefore Option(1) is correct.

To find Area of quadrilateral PMNR, we can calculate it by subtracting Area($\triangle$MNQ) from Area($\triangle$PQR):

Area($\triangle$PQR) $= \frac{1}{2}\times \| (0 \cdot (0+6) + 6 \cdot (-6+0) + 0 \cdot (0-0)) |$

Area($\triangle$PQR) $= \frac{1}{2}\times$ (6×6)

Area($\triangle$PQR) $= 18$

Similarily, Area($\triangle$MNQ) $= 2$

Therfore Area of Quadrilateral PMNR = Area($\triangle$PQR) - Area($\triangle$MNQ) = 18 - 2 = 16
Hence Option(7) is correct.

2. Let A be a set such that
A=$\{x|x^3 + 3x^2 - 61x - 63 = 0\ \}$
$B = \{x|x^3 - 2x^2 - 109x + 110 = 0\}$
$C = \{-1, -9, 1, 12, 15\}$
Let $D = (A \cap C) \cup (B \cap C)$

Suppose P(x) is a polynomial whose roots consist of all the elements of D and is of degree 3 and leading coefficient is $+1$. which of the following options is/are correct.

- ○ Number of Turning Points of P(x) is 2 ✓
- ○ Number of Turning Points of P(x) is 1 ✗
- ○ For interval $[-1, 1]$ the graph of $p(x)$ is first increasing and then decreasing ✗
- ○ For interval $[-1, 1]$ the graph of $p(x)$ is first decreasing and then increasing ✓

Solving Set A:

A=$\{x|x^3+3x^2-61x-63=0\ \}$
We can write $x^3+3x^2-61x-63=0$ as $(x-7)(x+1)(x+9)=0$ Therefore A can be written as $\{-9,-1,7\}$

Solving Set B:

$B=\{x|x^3-2x^2-109x+110=0\}$
We can write $x^3-2x^2-109x+110=0$ as $(x-11)(x-1)(x+10)=0$ Therefore B can be written as $\{-10,1,11\}$

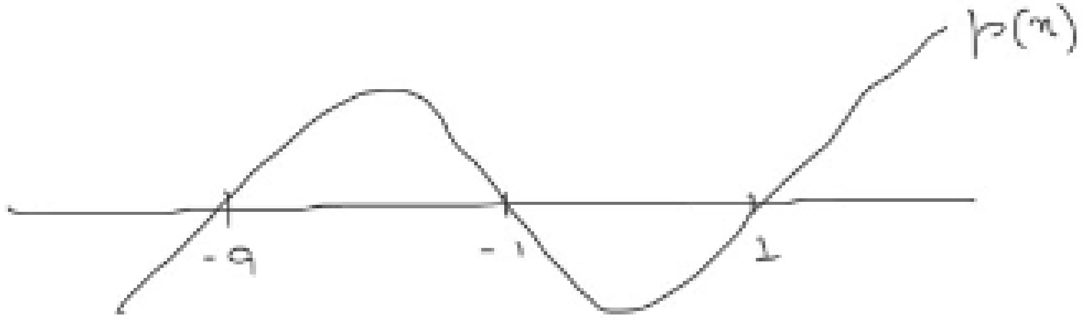
Hence, $(A \cap C) = \{-9, -1\}$ and $(B \cap C) = \{1\}$ Therefore, $D = (A \cap C) \cup (B \cap C)$
$D = \{-9, -1, 1\}$

Now, as given in question P(x) is a polynomial whose roots are elements of D. Hence the roots of P(x) is -9 , -1 and 1. Therefore we can write P(x) $= a \cdot (x + 9)(x + 1)(x - 1)$
Since leading coefficient is 1 therefore

$$P(x) = (x + 9)(x + 1)(x - 1)$$

$$P(x) = (x+9)(x+1)(x-1)$$

On plotting the graph of P(x):



We can clearly see that the number of Turning Points of Graph is 2 and The graph first decreases and then increases in the interval [-1 , 1] Therefore, Option(1) and Option(4) is correct.

If l and m are two distinct roots of quadratic equation $x^2 + lx + m = 0$, then find the value of l and m ?

[Hint: For a quadratic equation $ax^2 + bx + c$, Sum of roots $= \frac{-b}{a}$, Product of roots $= \frac{c}{a}$]

- ○ $l = 1$, $m = -2$
- ○ $l = 0$, $m = 1$
- ○ $l = -2$, $m = 0$
- ○ $l = -2$, $m = 1$

Solution:

Given a quadratic equation $x^2 + lx + m = 0$ with two distinct roots l and m.

Using the sum and product of roots of a quadratic equation, we have:
Sum of the roots is:

$$l + m = \frac{-b}{a} = -l$$
$$\implies 2l = -m$$
$$\implies l = \frac{-m}{2}$$

Product of the roots is:

$$lm = \frac{c}{a}$$
$$\implies lm = m$$
$$\implies l = 1$$

We know that $l = \frac{-m}{2}$, $l = 1$

$$\implies m = -2$$

$$\boxed{\text{l=1,m=-2}}$$

THANK YOU